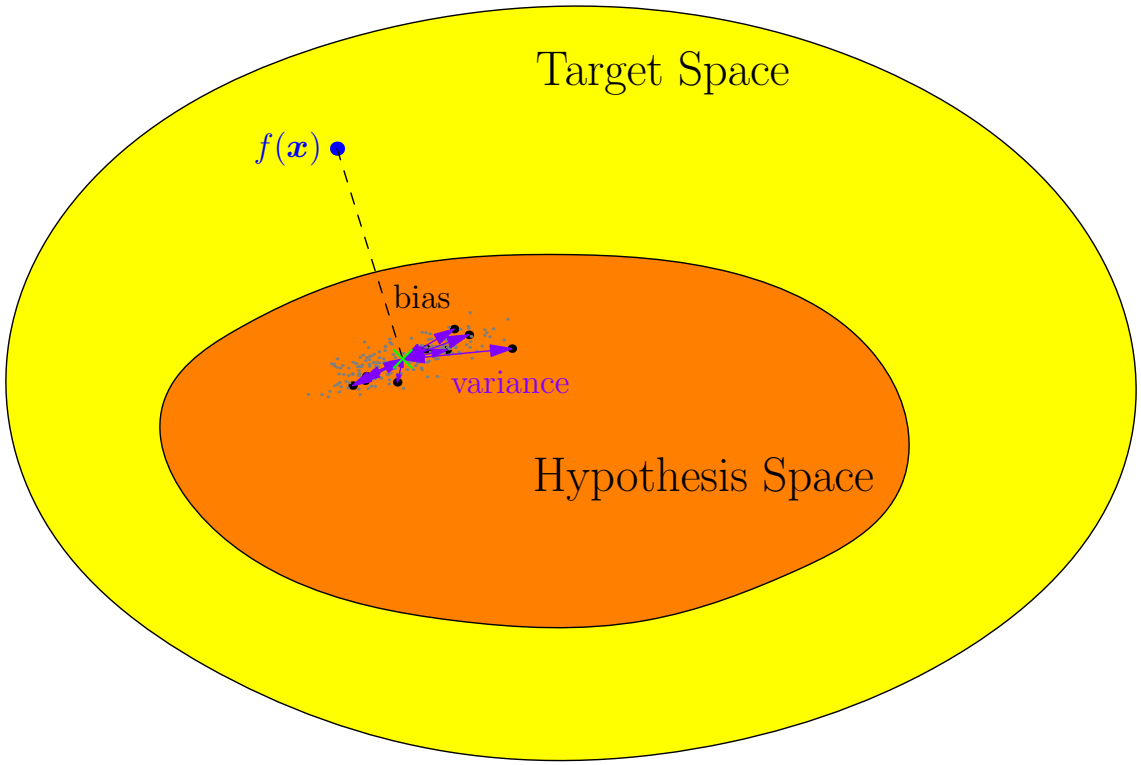
Mathematics of Machine Learning

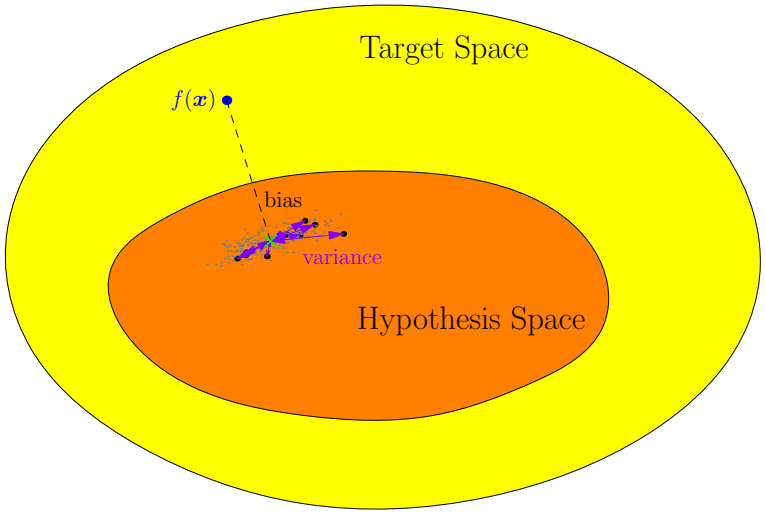
When Machine Learning Works



When ML Works, Bias Variance

Outline

1. **What Makes a Good Learning Machine?**
2. Bias-Variance Dilemma
3. Shrinkage and Ridge Regression
4. Ensembles



What Makes a Good Learning Machine?

- We want to understand why some machine learning techniques work well and others don't
- To understand why they work we need to understand what makes a good learning machine
- For this we have to get conceptual and think about **generalisation** performance

  generalisation: *how well do we do on unseen data as opposed to the training data*

What Makes Machine Learning Hard?

- Typically we work in high dimensions (i.e. have many features)
- The problem can be over-constrained (i.e. we have conflicting data to deal with)—solve by minimising an error function
- The problem can be under-constrained (i.e. there are many possible solutions that are consistent with the data)—need to choose a plausible solution
- Typically in machine learning the data will be over-constrained in some dimensions and under-constrained in others
- We can't visualise the data to know what is going on

Least Squares

- Suppose we want to learn some output y for a feature vector $\boldsymbol{x}$
- We construct a learning machine that makes a prediction $\hat{f}(\boldsymbol{x}|\boldsymbol{\theta})$
- We typically choose the machine to minimise a *training loss*

$$L_T(\mathcal{D}) = \sum_{(\boldsymbol{x},y)\in\mathcal{D}} \left(\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}) - y\right)^2 = \sum_{i=1}^{m} \left(\hat{f}(\boldsymbol{x}_i|\boldsymbol{\theta}) - y_i\right)^2$$

  where $\mathcal{D} = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^m$ is a set of size m, sampled from a probability distribution $\mu(\boldsymbol{x}, y)$

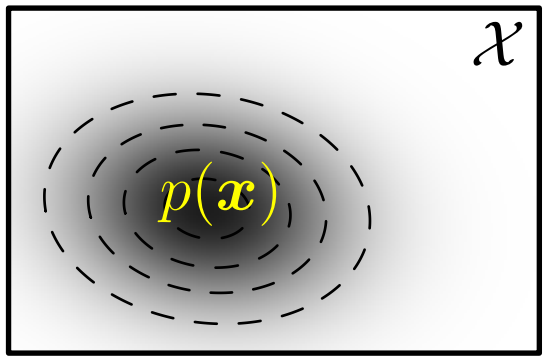
- We call this machine $\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{\mathcal{D}})$

Generalisation Error

- We want to minimise the *generalisation loss* which in this case is

$$L_G(\boldsymbol{\theta}_{\mathcal{D}}) = \sum_{(\boldsymbol{x},y)\in\mathcal{Z}} \mu(\boldsymbol{x},y)\left(\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{\mathcal{D}}) - y\right)^2$$

$$\mu(\boldsymbol{x},y) = p(y|\boldsymbol{x})p(\boldsymbol{x})$$

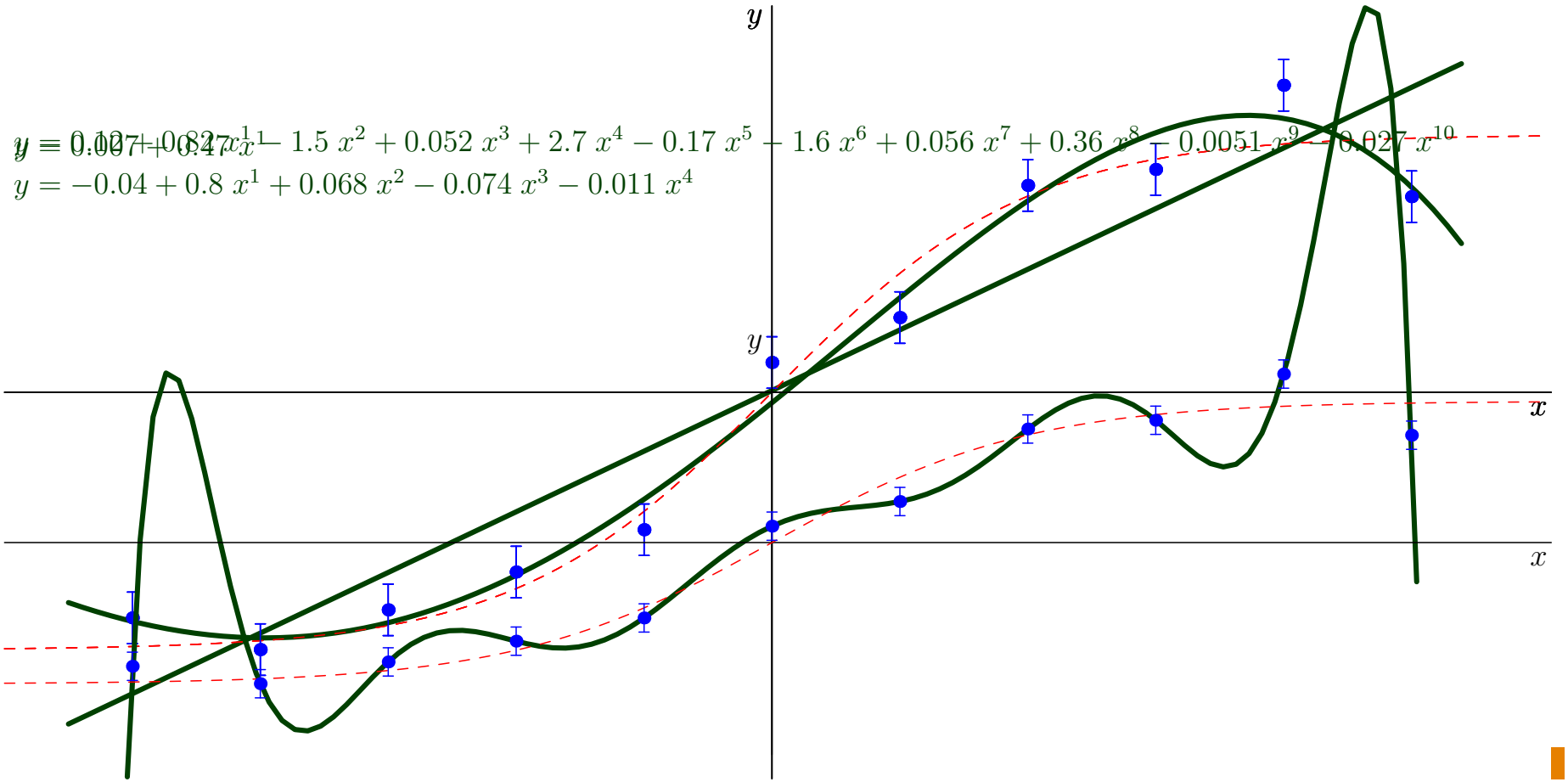


(we can estimate this if we have some labelled examples $(\boldsymbol{x}_i,y_i)$ which we have not trained on)

- We want to minimise $L_G(\boldsymbol{\theta}_{\mathcal{D}})$ but in practice we areb minimising $L_T(\mathcal{D})$, *what could possibly go wrong?*

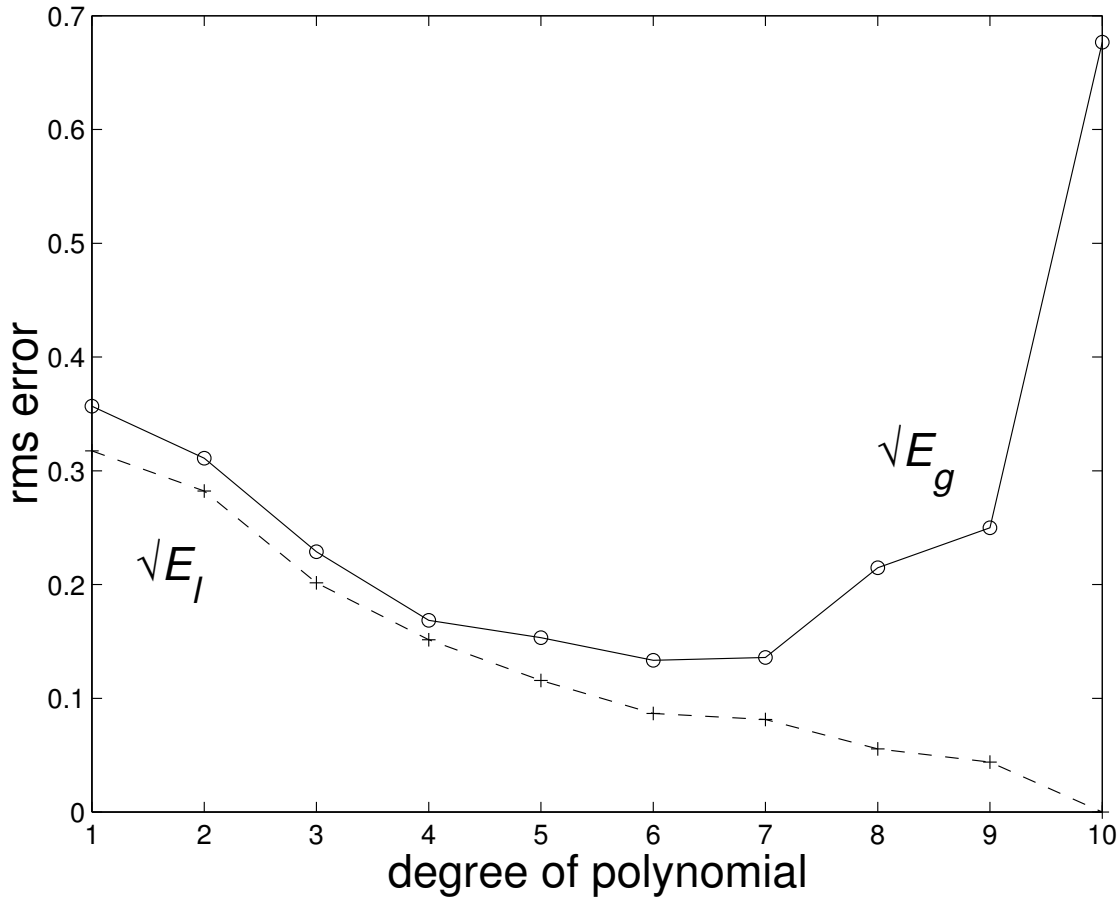
Too Simple or Too Complex?

- Fit $\hat{f}(x|\boldsymbol{\theta}_{\mathcal{D}})$ to data



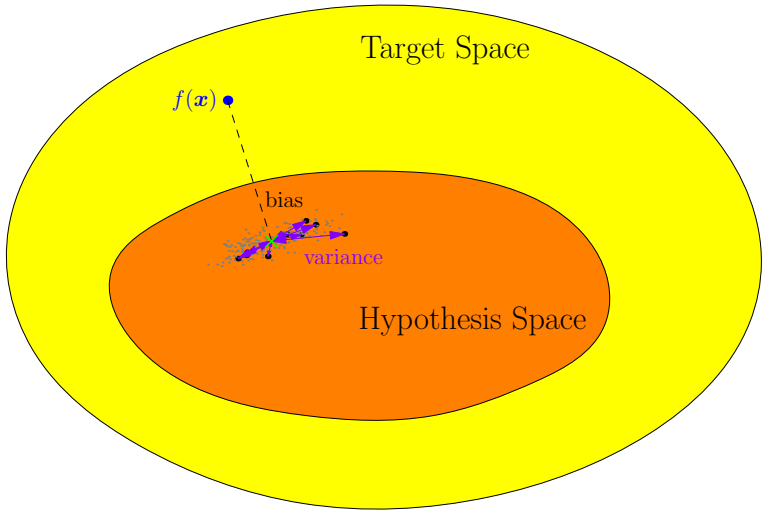
Measuring Generalisation Error for Regression

- Consider the regression example. The root mean squared error is



Outline

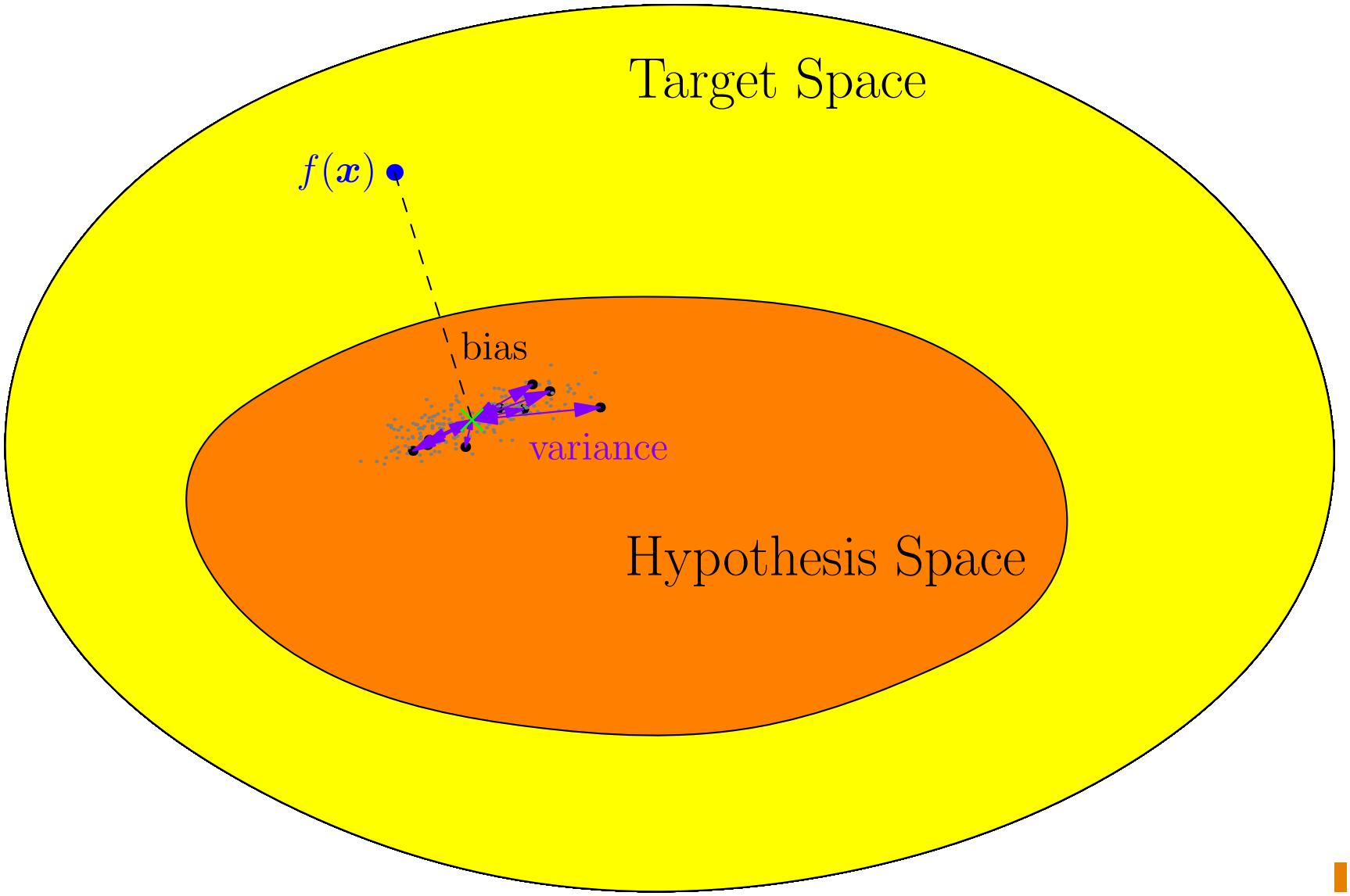
1. What Makes a Good Learning Machine?
2. **Bias-Variance Dilemma**
3. Shrinkage and Ridge Regression
4. Ensembles



Expected Generalisation Performance

- Our generalisation performance will depend on our training set, $\mathcal{D}$
- To reason about generalisation we can ask what is the $expected\ generalisation\ loss$, when we average over all different data sets of size m drawn independently from $\mu(\boldsymbol{x},y)$
- For each data set, $\mathcal{D}$, we would learn a different approximator $\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{\mathcal{D}})$
- Note that in practice we only get one data set. We might be lucky and do better than the expected generalisation or we might be unlucky and do worse

Approximation and Estimation Errors



Mean Machine

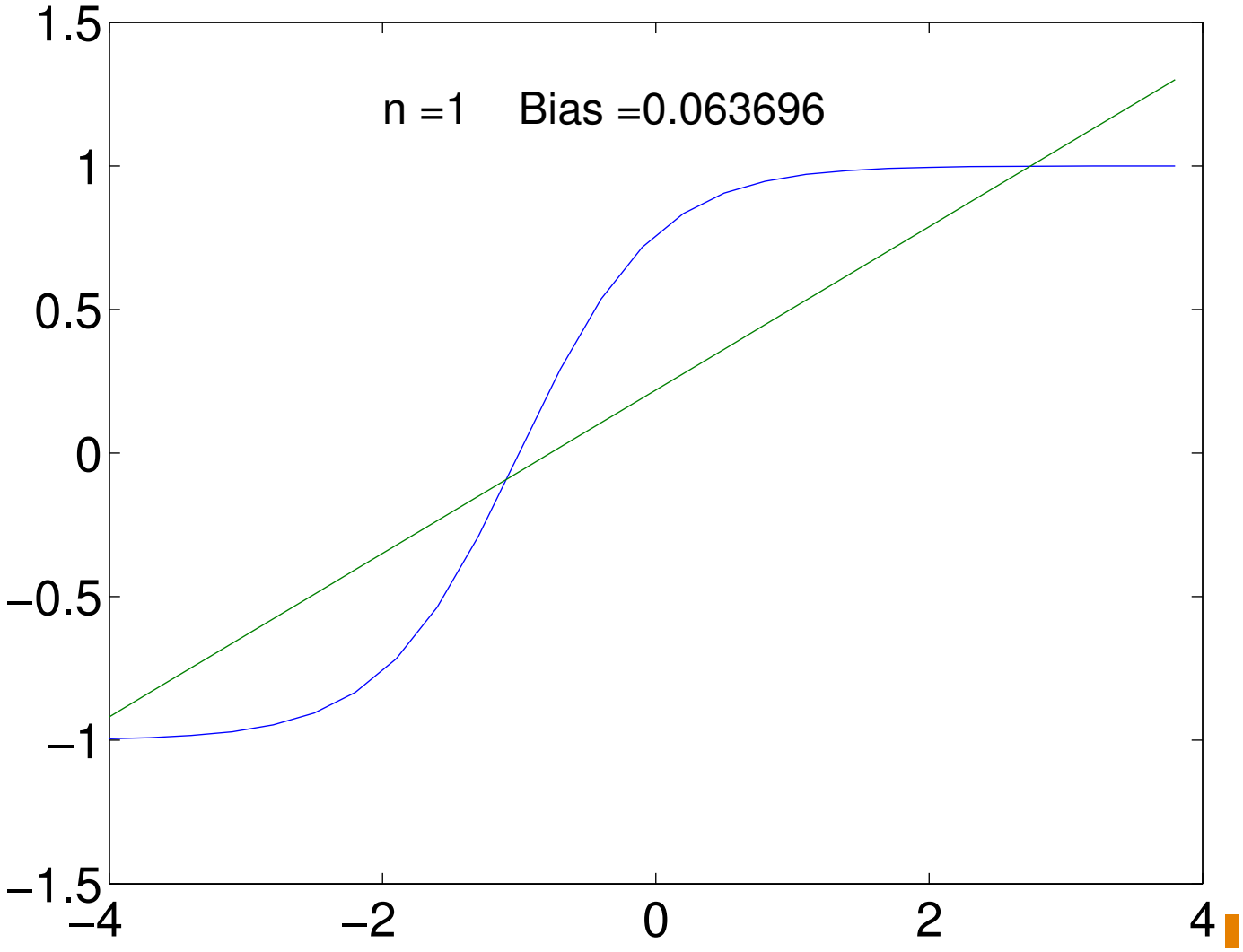
- To help understand generalisation we can consider the mean prediction with respect to machines trained with all data sets of size m

$$\hat{f}_m(\boldsymbol{x}) = \mathbb{E}_{\mathcal{D}}\left[\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{\mathcal{D}})\right]$$

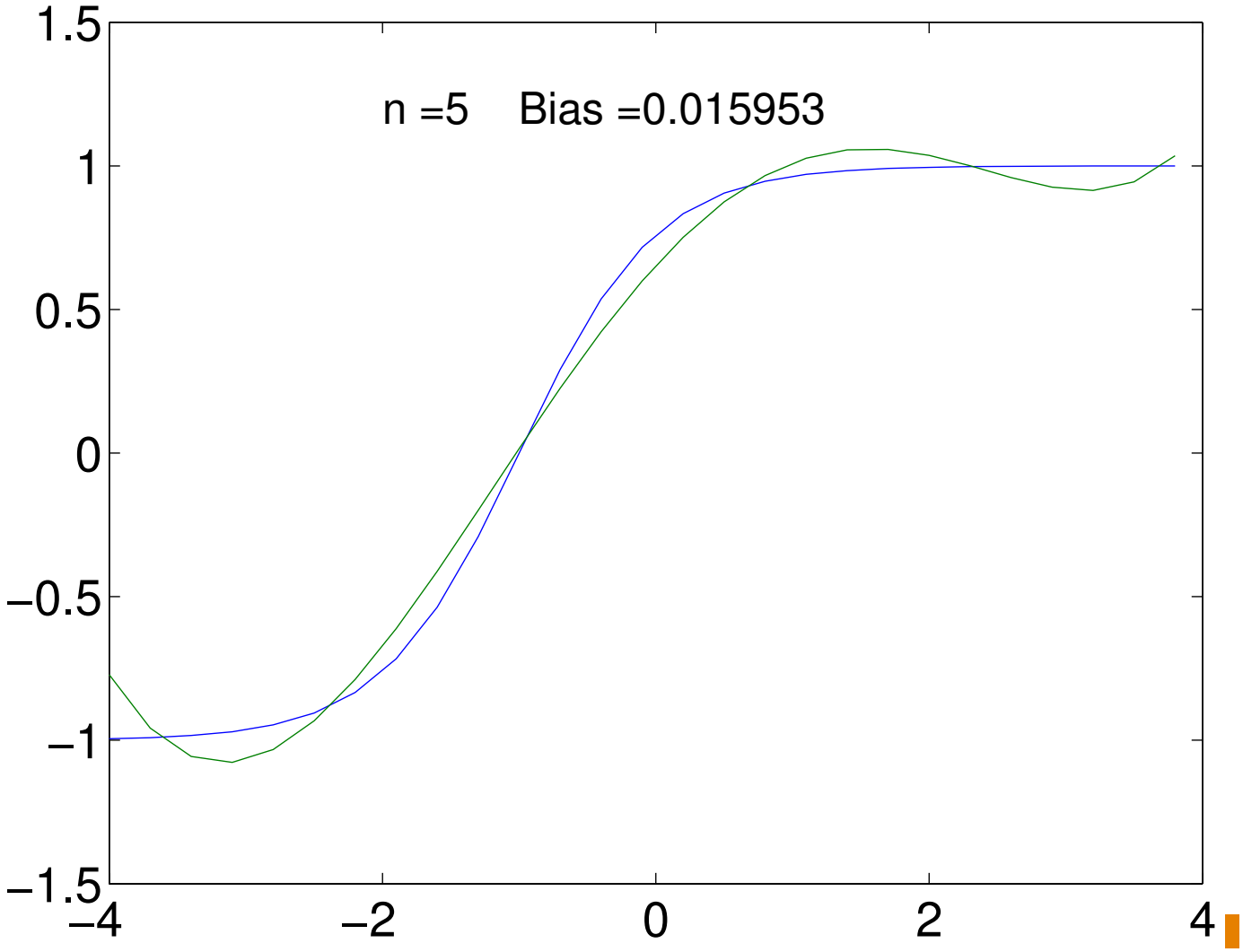
- We can define the **bias** to be generalisation performance of the mean machine

$$B = \sum_{(\boldsymbol{x},y)} \mu(\boldsymbol{x},y)\left(\hat{f}_m(\boldsymbol{x}) - y\right)^2$$

Regression Example $n = 1$



Regression Example $n = 5$



Bias and Variance

Consider the expected generalisation for data sets of size $|\mathcal{D}| = m$

$$\bar{L}_G = \mathbb{E}_{\mathcal{D}}[L_G(\boldsymbol{\theta}_{\mathcal{D}})] = \mathbb{E}_{\mathcal{D}}\left[\sum_{(\boldsymbol{x},y)} \mu(\boldsymbol{x},y)\left(\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{\mathcal{D}}) - y\right)^2\right]$$

$$= \sum_{(\boldsymbol{x},y)} \mu(\boldsymbol{x},y)\mathbb{E}_{\mathcal{D}}\left[\left(\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{\mathcal{D}}) - y\right)^2\right]$$

$$= \sum_{(\boldsymbol{x},y)} \mu(\boldsymbol{x},y)\mathbb{E}_{\mathcal{D}}\left[\left(\left(\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{\mathcal{D}}) - \hat{f}_m(\boldsymbol{x})\right) + \left(\hat{f}_m(\boldsymbol{x}) - y\right)\right)^2\right]$$

$$= \sum_{(\boldsymbol{x},y)} \mu(\boldsymbol{x},y)\left(\mathbb{E}_{\mathcal{D}}\left[\left(\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{\mathcal{D}}) - \hat{f}_m(\boldsymbol{x})\right)^2 + \left(\hat{f}_m(\boldsymbol{x}) - y\right)^2\right]\right.$$

$$\left. + 2\mathbb{E}_{\mathcal{D}}\left[\left(\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{\mathcal{D}}) - \hat{f}_m(\boldsymbol{x})\right)\left(\hat{f}_m(\boldsymbol{x}) - y\right)\right]\right)$$

Cross Term

- The cross term vanishes

$$
\begin{aligned}
C &= \mathbb{E}_{\mathcal{D}}\left[\left(\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{\mathcal{D}}) - \hat{f}_m(\boldsymbol{x})\right)\left(\hat{f}_m(\boldsymbol{x}) - y\right)\right] \\
&= \mathbb{E}_{\mathcal{D}}\left[\left(\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{\mathcal{D}}) - \hat{f}_m(\boldsymbol{x})\right)\right]\left(\hat{f}_m(\boldsymbol{x}) - y\right) \\
&= \left(\mathbb{E}_{\mathcal{D}}\left[\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{\mathcal{D}})\right] - \hat{f}_m(\boldsymbol{x})\right)\left(\hat{f}_m(\boldsymbol{x}) - y\right) \\
&= \left(\hat{f}_m(\boldsymbol{x}) - \hat{f}_m(\boldsymbol{x})\right)\left(\hat{f}_m(\boldsymbol{x}) - y\right) = 0
\end{aligned}
$$

- Thus

$$
\bar{L}_G = \sum_{(\boldsymbol{x},y)} \mu(\boldsymbol{x},y)\mathbb{E}_{\mathcal{D}}\left[\left(\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{\mathcal{D}}) - \hat{f}_m(\boldsymbol{x})\right)^2 + \left(\hat{f}_m(\boldsymbol{x}) - y\right)^2\right]
$$

Bias and Variance

- We can write the expected generalisation loss as

$$\mathbb{E}_{\mathcal{D}}[L_G(\boldsymbol{\theta}_{\mathcal{D}})] = \sum_{(\boldsymbol{x},y)} \mu(\boldsymbol{x},y)\mathbb{E}_{\mathcal{D}}\left[\left(\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{\mathcal{D}}) - \hat{f}_m(\boldsymbol{x})\right)^2\right] + \sum_{(\boldsymbol{x},y)} \mu(\boldsymbol{x},y)\left(\hat{f}_m(\boldsymbol{x}) - y\right)^2 = V + B$$

- Where B is the bias and V is the variance defined by

$$V = \sum_{(\boldsymbol{x},y)} \mu(\boldsymbol{x},y)\mathbb{E}_{\mathcal{D}}\left[\left(\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{\mathcal{D}}) - \hat{f}_m(\boldsymbol{x})\right)^2\right]$$

Where Is the Noise?

- Suppose $y = f^*(\boldsymbol{x}) + \epsilon$ where the noise ϵ has mean 0 and variance σ^2
- Then (the cross term vanishes again, as $\mathbb{E}[\epsilon] = 0$)

$$B = \sum_{(\boldsymbol{x},y)} \mu(\boldsymbol{x},y)\big(\hat{f}_m(\boldsymbol{x}) - y\big)^2 = \sigma^2 + \sum_{\boldsymbol{x}} \mu(\boldsymbol{x})\big(\hat{f}_m(\boldsymbol{x}) - f^*(\boldsymbol{x})\big)^2$$

- So $\bar{L}_G =$ noise $+$ bias^2 $+$ variance, the form you may have seen elsewhere
- The noise is the same for every machine: we can do nothing about it
- Here we keep it inside B: $\bar{L}_G = B + V$

Bias-Variance Dilemma

- The bias measures the generalisation performance of the $mean\ machine$ and is large if the machine is too simple to capture the changes in the function we want to learn
- The variance measures the variation in the prediction of the machines as we change the data set we train on

$$V = \sum_{(\boldsymbol{x},y)} \mu(\boldsymbol{x},y) \mathbb{E}_{\mathcal{D}} \left[\left(\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{\mathcal{D}}) - \hat{f}_m(\boldsymbol{x}) \right)^2 \right]$$

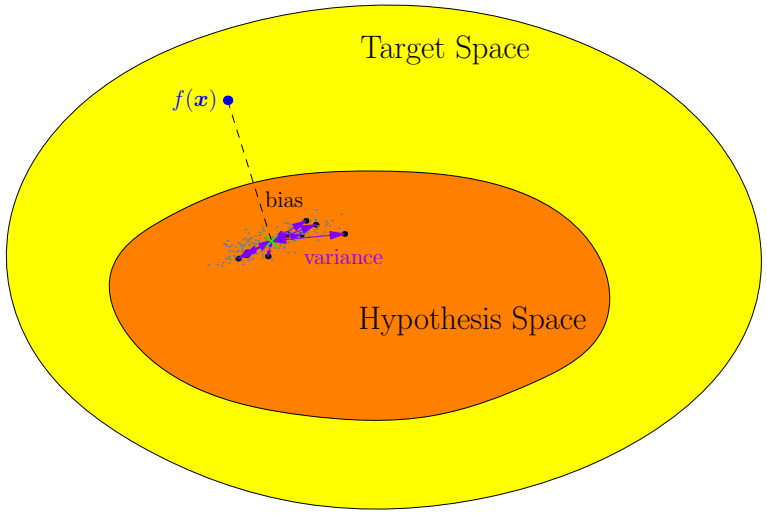
- The variance is usually large if we have a complex machine
- Striking the right balance is often the key to getting good results

Balancing Bias and Variance

- We want to choose a learning machine that is complex enough to capture the underlying function we are trying to learn, but otherwise as simple as possible
- There are a number of tricks to achieve this balance
- Some require us to preprocess the data to reduce the number of inputs
- Some machines cleverly adjust their own complexity
- The rest of this lecture: two exact examples, and averaging

Outline

1. What Makes a Good Learning Machine?
2. Bias-Variance Dilemma
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Shrinking an Estimate

- The simplest learning problem: estimate the mean θ of a distribution with variance σ^2 from m samples
- The sample mean $\bar{x}$ is unbiased with variance σ^2/m
- Try the deliberately biased $\hat{\theta} = c\bar{x}$ with $0 \leq c \leq 1$

$$\mathbb{E}\left[\left(\hat{\theta} - \theta\right)^2\right] = \underbrace{(1-c)^2\theta^2}_{\text{bias}^2} + \underbrace{c^2\frac{\sigma^2}{m}}_{\text{variance}}$$

- At $c = 1$ the derivative is $2\sigma^2/m > 0$: shrinking a little **always** reduces the error
- The best c depends on the unknown θ, so in practice we choose it by validation

Ridge Regression

- Data $\boldsymbol{y} = \mathbf{X}\boldsymbol{w}^* + \boldsymbol{\epsilon}$, with noise of variance σ^2 on each target
- Ridge regression minimises $\|\mathbf{X}\boldsymbol{w} - \boldsymbol{y}\|^2 + \eta\|\boldsymbol{w}\|^2$, giving

$$\hat{\boldsymbol{w}} = \left(\mathbf{X}^\mathsf{T}\mathbf{X} + \eta\mathbf{I}\right)^{-1}\mathbf{X}^\mathsf{T}\boldsymbol{y}$$

- $\eta = 0$ is least squares: unbiased, but how variable?
- Use the SVD $\mathbf{X} = \mathbf{U}\boldsymbol{S}\boldsymbol{V}^\mathsf{T}$ and work in the basis of the columns of $\boldsymbol{V}$, where everything separates

Ridge Through the SVD

- Write $w_k^* = \boldsymbol{v}_k^\top \boldsymbol{w}^*$ for the component of $\boldsymbol{w}^*$ along $\boldsymbol{v}_k$; since $\boldsymbol{u}_k^\top \boldsymbol{y} = s_k w_k^* + \boldsymbol{u}_k^\top \boldsymbol{\epsilon}$, the same component of the estimate is

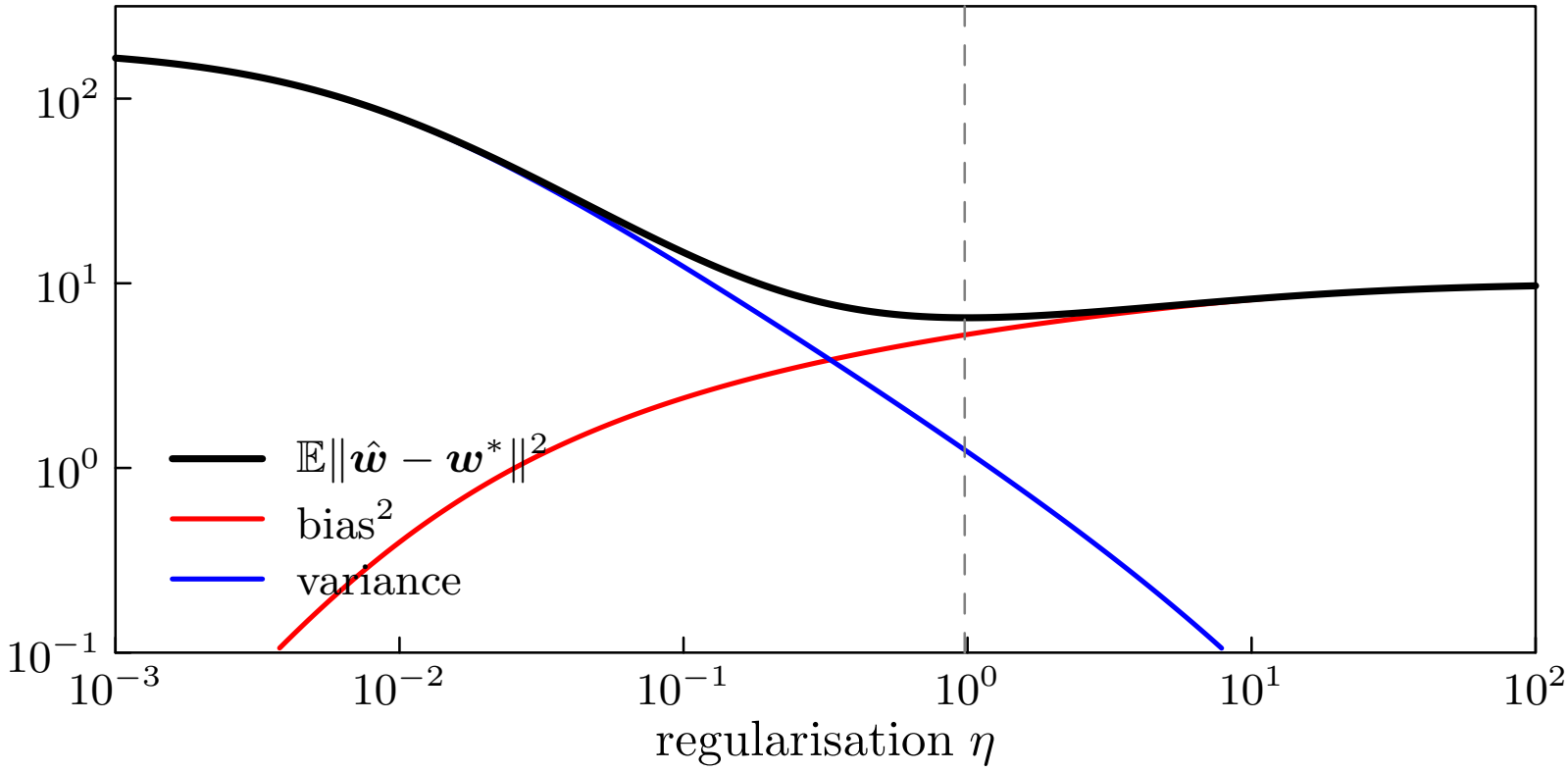
$$\hat{w}_k = \frac{s_k \boldsymbol{u}_k^\top \boldsymbol{y}}{s_k^2 + \eta}, \qquad \mathbb{E}[\hat{w}_k] = \frac{s_k^2}{s_k^2 + \eta} w_k^*, \qquad \mathrm{Var}(\hat{w}_k) = \frac{\sigma^2 s_k^2}{(s_k^2 + \eta)^2}$$

- So the expected squared error of the weights is

$$\mathbb{E}\left[\|\hat{\boldsymbol{w}} - \boldsymbol{w}^*\|^2\right] = \sum_k \underbrace{\left(\frac{\eta}{s_k^2 + \eta}\right)^2 w_k^{*2}}_{\text{bias}^2} + \underbrace{\frac{\sigma^2 s_k^2}{(s_k^2 + \eta)^2}}_{\text{variance}}$$

- With $\eta = 0$ the variance is $\sum_k \sigma^2 / s_k^2$: small singular values (ill-conditioning) make it huge; η shrinks exactly those directions, trading a little bias for a large drop in variance

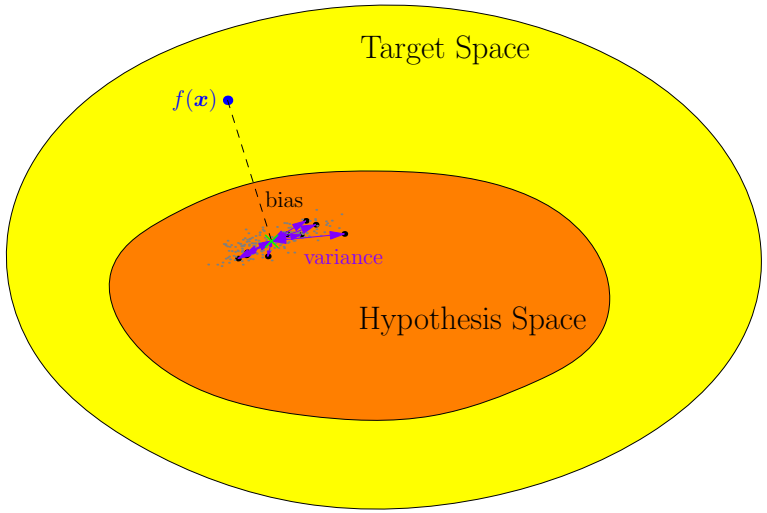
Choosing η



- Ten singular values from 3 down to 0.1, $\sigma = 1$: least squares has error 188, the best $\eta \approx 1$ gives 6.5
- In practice choose η by cross-validation

Outline

1. What Makes a Good Learning Machine?
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Averaging Reduces Variance

- An **ensemble** averages M predictors, $\bar{f}_M(\boldsymbol{x}) = \frac{1}{M}\sum_{i=1}^{M} f_i(\boldsymbol{x})$, trained with different randomness
- Averaging does not change the mean, so the bias is unchanged
- If each $f_i(\boldsymbol{x})$ has variance s^2 and each pair has correlation ρ

$$\mathrm{Var}\big(\bar{f}_M(\boldsymbol{x})\big) = \rho s^2 + \frac{1-\rho}{M} s^2$$

- More members remove the second term, but correlation sets a floor ρs^2
- So we want members that are good **and** different

Bagging, Forests and Deep Ensembles

- **Bagging**: train each member on a bootstrap resample of the data (sampled with replacement)
- **Random forests**: bagged decision trees that also choose each split from a random subset of the features—each tree is worse, but the trees are less correlated
- **Deep ensembles**: train several networks on the same data from different random initialisations
- The spread of the members' predictions is a useful measure of uncertainty
- Ensembles of high-variance, low-bias machines are among the most reliable methods in practice

Lessons

- The expected generalisation loss splits into bias plus variance
- One of the most useful intuitions about what works is the bias-variance framework
- A biased estimate can beat an unbiased one: shrinkage, ridge regression
- The bias is high for simple machines that can't capture the data
- The variance is high for complex machines that are sensitive to the training set
- Averaging (ensembles) reduces variance without adding bias